# Projective Gram Recurrence for the Normalized Directional Base

## A Sharp Sufficient Packet and Its Finite Polar-Gauge Audit

Prime Dynamics Theory Program

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**Abstract**

The remaining directional support frontier contains the normalized Gram base

$$a(G) = \sqrt{\lambda_{\min}(G)/\lambda_{\max}(G)}.$$

We give a sharp multiplicative recurrence for this quantity. If $G, H$ are positive definite and $\mu_{\min}, \mu_{\max}$ are the generalized eigenvalue extrema of $H$ relative to $G$, then the Hilbert projective distance

$$d_{\text{proj}}(G, H) = \log(\mu_{\max}/\mu_{\min})$$

satisfies

$$a(H) \geq e^{-d_{\text{proj}}(G,H)/2} a(G).$$

The coefficient 1/2 is exact. Consequently, summable projective variation of an aligned Gram chain implies a positive normalized-base liminf. Merely bounded one-step distortion does not: $G_n = \operatorname{diag}(e^n, 1)$ has $d_{\text{proj}}(G_n, G_{n+1}) = 1$ while $a(G_n) \to 0$.

We audit the theorem on all 330 RH-135 Gram transitions using the physical polar alignment of the packet frames. Every one-step and cumulative bound holds. The numerical verdict is nevertheless negative for this direct sufficient route: step distances range from 3.92 to 58.68, chain totals from 97.53 to 234.16, and the resulting terminal product lower ranges from $1.66 \times 10^{-52}$ to $3.16 \times 10^{-23}$. Exact terminal bases are between $1.04 \times 10^{-10}$ and $5.11 \times 10^{-3}$, so the universal projective product discards enormous correlation. The theorem closes a rigorous logical subproblem, but the finite archive neither proves nor supports all-level projective summability.


# 1 The base packet

RH-139 reduces eventual directional support to two exact-matrix conditions: a controlled squared-tail envelope $y_n$ with $\limsup y_n < 1$ and a normalized base $a_n$ with $\liminf a_n > 0$. RH-140–RH-145 strengthen the finite enclosure and viability side. The second condition has so far been read directly from each Gramian. The purpose of this paper is to express it as a recurrence between consecutive Gramians.

For $G, H \succ 0$, let $\mu_{\min}$ and $\mu_{\max}$ denote the smallest and largest eigenvalues of $G^{-1/2}HG^{-1/2}$. Equivalently,

$$\mu_{\min} G \preceq H \preceq \mu_{\max} G. \tag{1}$$

The quantity

$$d_{\mathrm{proj}}(G, H) = \log \frac{\mu_{\max}}{\mu_{\min}}$$

is the Hilbert projective distance on the positive definite cone [Birkhoff, 1957, Bushell, 1973, Bhatia, 2007]. It ignores common scalar rescaling and measures only shape distortion.

# 2 Sharp projective recurrence

**Theorem 1** (Projective normalized-base recurrence)**.** *For every pair $G, H \succ 0$,*

$$a(H) \geq e^{-d_{\mathrm{proj}}(G,H)/2} a(G). \tag{2}$$

*The factor $1/2$ in the exponent cannot be improved uniformly.*

*Proof.* The Loewner bounds (1) imply

$$\lambda_{\min}(H) \geq \mu_{\min} \lambda_{\min}(G), \qquad \lambda_{\max}(H) \leq \mu_{\max} \lambda_{\max}(G).$$

Taking their ratio and square root gives

$$a(H) \geq \sqrt{\frac{\mu_{\min}}{\mu_{\max}}} \sqrt{\frac{\lambda_{\min}(G)}{\lambda_{\max}(G)}} = e^{-d_{\mathrm{proj}}(G,H)/2} a(G).$$

For sharpness take $G = I_2$ and $H = \mathrm{diag}(e^d, 1)$. Then $d_{\mathrm{proj}}(G, H) = d$, $a(G) = 1$, and $a(H) = e^{-d/2}$, so equality holds. □

At transition $n$, the packet construction supplies an orthogonal polar alignment $O_n$. Apply Theorem 1 to $O_n^T G_n O_n$ and $G_{n+1}$. Since $a(O_n^T G_n O_n) = a(G_n)$, no base term is changed by the gauge.

**Corollary 2** (Summable projective variation)**.** *Let $G_n \succ 0$, let $O_n$ be any orthogonal alignments, and put*

$$d_n = d_{\mathrm{proj}}(O_n^T G_n O_n, G_{n+1}).$$

*Then for $n > N$,*

$$a(G_n) \geq a(G_N) \exp\left( -\frac{1}{2} \sum_{j=N}^{n-1} d_j \right). \tag{3}$$

*In particular, if $\sum_{j=N}^{\infty} d_j < \infty$, then*

$$\inf_{n \geq N} a(G_n) \geq a(G_N) e^{-\frac{1}{2} \sum_{j=N}^{\infty} d_j} > 0.$$

*Proof.* Iterate (2). Orthogonal conjugation preserves the extreme eigenvalues of the source Gramian. □

This gives a complete sufficient packet for the RH-139 base condition. It does not claim necessity: exact bases can stay positive despite infinite projective path length, for example under oscillation between two fixed shapes.

| $\sigma$ | median step $d_{\mathrm{proj}}$ | median chain variation | maximum variation | minimum exact terminal base |
|---|---|---|---|---|
| 0.16 | 44.75 | 135.60 | 142.05 | $1.04 \times 10^{-10}$ |
| 0.08 | 21.67 | 108.56 | 122.76 | $5.66 \times 10^{-9}$ |
| 0.04 | 14.24 | 150.28 | 188.46 | $8.22 \times 10^{-8}$ |
| 0.02 | 8.84 | 167.52 | 204.04 | $9.76 \times 10^{-4}$ |
| 0.01 | 10.81 | 228.01 | 234.16 | $2.14 \times 10^{-3}$ |

Table 1: Projective Gram audit by frozen scale anchor.

# 3 Why bounded distortion is insufficient

The natural weakening $\sup_n d_n < \infty$ does not control accumulated anisotropy.

**Proposition 3** (Nonsummable bounded-distortion obstruction)**.** *There is an SPD chain with $d_{\mathrm{proj}}(G_n, G_{n+1}) = 1$ for every $n$ but $a(G_n) \to 0$. Thus no positive universal base liminf follows from bounded one-step projective distortion alone.*

*Proof.* Take $G_n = \mathrm{diag}(e^n, 1)$. The relative matrix is $G_n^{-1/2} G_{n+1} G_n^{-1/2} = \mathrm{diag}(e, 1)$, hence every step has distance one. Yet $a(G_n) = e^{-n/2} \to 0$. Equality holds in the cumulative bound (3) at every level. □

The obstruction separates two statements that can otherwise be confused. Uniform local control prevents a single catastrophic jump; summability prevents infinitely many moderate jumps from building unbounded condition number. A block cancellation theorem or a signed shape-drift argument could replace summability, but such extra structure must be proved explicitly.

# 4 Physical polar-gauge audit

We rebuild the RH-135 records at 80 decimal digits. For consecutive packet input frames $U_n, U_{n+1}$, the physical gauge is the orthogonal polar factor of $U_n^T U_{n+1}$. The audit then computes the generalized eigenvalues of $G_{n+1}$ relative to $O_n^T G_n O_n$, the exact normalized bases, the one-step lower, and the cumulative product lower.

All 330 one-step inequalities and all 330 cumulative inequalities pass. The finite distribution is summarized in Table 1. Repeated rows at finer anchors occur because all three threshold branches coincide there.

The smallest observed step distance is 3.9206, not a small perturbative increment. Total variation grows again on the two longest chains, reaching 234.16. Consequently the direct product estimate is between many orders and nearly fifty orders below the exact terminal base. This is not a defect in Theorem 1: diagonal examples attain it exactly. It means the frozen Gram path contains favorable correlations that a product of universal one-step worst cases cannot retain.

The exact terminal bases themselves improve dramatically at the two finest anchors. That observation motivates a correlated base-tail state rather than an independent projective product. It is finite evidence only; five anchors cannot establish a liminf.

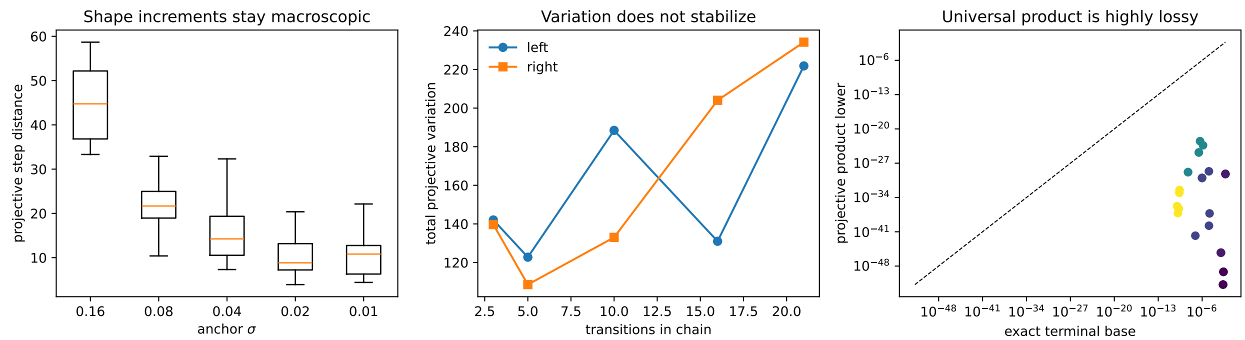


Figure 1: Step-distance distributions, accumulated projective variation, and the gap between exact terminal bases and universal product lowers.

# 5 Consequence and claim boundary

The normalized-base problem now has a rigorous fallback route:

$$\boxed{\sum_n d_{\mathrm{proj}}(O_n^T G_n O_n, G_{n+1}) < \infty} \quad \Longrightarrow \quad \boxed{\liminf_n a(G_n) > 0}.$$

The implication is exact, dimension independent, and sharp at the level of universal constants. The present data do not support the antecedent: the observed increments remain macroscopic and cumulative variation does not stabilize. Nor do finite data disprove it for a different asymptotic assembly, gauge, block grouping, or analytically normalized Gram process.

RH-146 does not prove projective summability for the intended model, a positive all-level normalized-base liminf, the controlled tail gap, Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis. The next step is to preserve the observed correlation between base recovery and tail viability instead of multiplying their separate worst-case bounds.

# References

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